

Figure 10: Non-shared lock—mandatory file locking

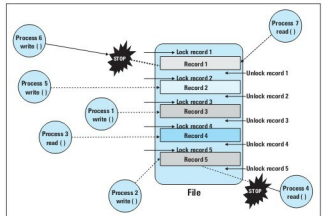


Figure 12: Shared lock—advisory or record locking

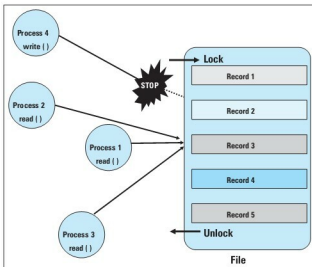


Figure 11: Shared lock—mandatory file locking

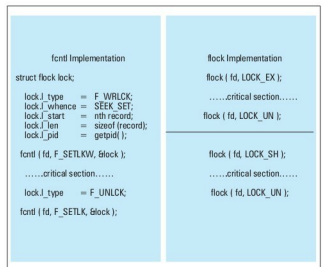


Figure 13: Sketch of file-locking implementation

Let's look at the members of the struct:

- *l_type* specifies the lock type, and is one of the following: read lock (*F_RDLCK*), write lock (*F_WRLCK*) or unlock a locked file (*F_UNLCK*). A read lock will allow many readers to access the file at the same time, but no writer is allowed. A write lock will allow only one writer at a time.
- 1. *l_whence* is used to set the file pointer relative to the current cursor position. If, for example, you want to go to the starting location of a file, then pass *SEEK_SET*, or if the file is already open and you subsequently want to set the lock from the current cursor position, then pass *SEEK_CUR*. If you want to go to the end of the file, then pass *SEEK_END*.
- 2. *l_start* member of the structure is used to tell the starting point of the file, which could be from the 10th byte.
- 3. *l_len* is used to specify till what point you want to lock the file. It could be till the 100th byte.
- *l_pid* is the process ID of the process that holds the lock.

If the process that is holding the lock exits abnormally, then the kernel will release the lock, and the next deserving process can acquire the lock.

If we want to lock the entire file, we set the *l_start* and *l_len* fields to 0. To improve locking granularity, however, each process can lock a specific record, or range of bytes in a file, by setting appropriate values in the *l_start* and *l_len* fields. Figure 13 explains how to approach this design of locking a specific record or byte, by the suitable setting of the *l_start* and *l_len* fields.

The locking and unlocking is done by calling the *fcntl* system call. Its syntax is:

```
int fcntl (int fd, int cmd, struct flock *lock);
```

We have three commands: *F_SETLKW*, *F_SETLK* and *F_GETLK* to acquire, release or check the availability of the lock. The first two commands are used to acquire or release the lock, but the difference between them is that